Name: ______________________________ Roll: ___________________________

CSE 556: Natural Language Processing (NLP) — QUIZ - 2

Date: 8th Apr 2025 **Duration:** 30 mins. **Max Marks: 20**

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1. Define Emotion-Cause extraction pair. Also, identify all instances of emotion-cause pairs in the following paragraph. **[3]**

   “Yesterday morning, a policeman visited the old man with the lost money, and told him that the thief was caught. The old man was very happy, and deposited the money in the bank.”

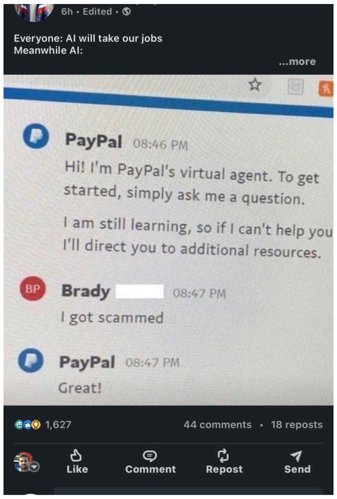
Extracting the potential causes behind expressed emotions

Emotion-Cause Pair Extraction (ECPE)

(The old man was very happy, a policeman visited the old man with the lost money)

(The old man was very happy, and told him that the thief was caught)

2. Consider the following snapshot and identify the reason behind Paypal's inappropriate response in the third utterance (PayPal: Great!) **[2]**



6h • Edited •

Everyone: AI will take our jobs
Meanwhile AI:

...more

PayPal 08:46 PM
Hi! I'm PayPal's virtual agent. To get started, simply ask me a question.

I am still learning, so if I can't help you I'll direct you to additional resources.

Brady [redacted] 08:47 PM
I got scammed

PayPal 08:47 PM
Great!

Agent/Bot could not understand/comprehend the requirement or intent of the user’s utterance.

(Anything on similar lines).

3. Identify the emotions of the following characters: **[0.5 * 9 + 0.5 = 5]**

   Bonus of 0.5 marks only if all emotions are predicted correctly.

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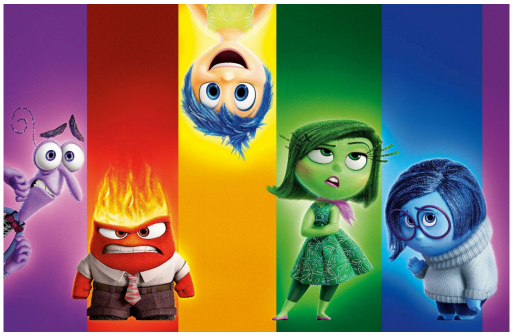

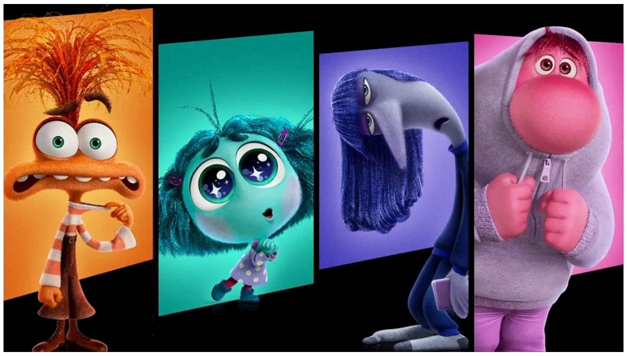

a. Fear b. Angry c. Joy/Happy d. Disgust e. Sad f. Anxiety g. Envy h. Ennui/Bore i. Embarrass

4. Define your NLP course project considering the following details: **[7]**

Project Title (1 line) – 1 mark:	
Number of samples in the test set (Just number) – 2 marks:	We need numbers (Approximately within +/- 5% of the test samples). Percentage will not work.
Name of evaluation metric used (Just name) – 2 marks:	
Clear Problem statement (2-3 lines) – 2 marks: **As per your problem.**	

5. What are the pros and cons of reactive and proactive combating of hate speeches? Mention at least one for each case. **[2]**

Reactive:	Proactive:
● Pros: Freedom of speech is not compromised	● Pros: Tries to handle the situation before a potential damage can happen.
● Cons: Damage is already done.	● Cons: Freedom of speech can be compromised. OR It's suggestive in nature – users may still ignore it and there will be no combating.

6. For any generative task, which kind of evaluation is most important and why? **[1]**

Linguistic-based evaluation metrics for generative text such BLEU, Rouge, METEOR, etc. are too rigid and forces a direct match between the generated text and the reference text. In many cases, two texts with the same semantics can be completely different. Semantic , Domain Expert or Human Evaluation check the quality of output at the meaning-level. “This movie is good.”, “This movie is not good.” ⇒ Opposite semantics but BLEU score will be high. “This movie is good.”, “What a good movie is this?” ⇒ Same semantic but BLEU score will be low

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